

(RAY) REWAT KHANTHAPORN

AI-Focused Data Analytics | Gen AI | AI Agents | Decision Intelligence

Auckland, New Zealand
linkedin.com/in/ray-rewat-khanthaporn
Email: available on request
New Zealand Permanent Resident

GEN AI | AI AGENTS | SNOWFLAKE

Machine Learning | Applied AI/NLP | Python | PyTorch | scikit-learn | Transformers | R | SQL | dbt | Qlik | Automation | Predictive Modelling

PROFESSIONAL PROFILE

AI-focused data analytics professional with experience in applied AI, machine learning, statistical modelling, analytics engineering and audit assurance. Strong capability in Snowflake, SQL, R and Python, with practical exposure to Gen AI concepts, AI architecture and applied AI/NLP tools. Experienced in building reproducible workflows, automated exception-testing frameworks, validated analytical outputs and stakeholder-ready dashboards across healthcare, audit assurance, finance and research environments. Completed postgraduate Artificial Intelligence coursework at the University of Auckland, including AI Fundamentals and AI Architecture and Design, with AI coursework GPA of 6.5. Skilled in translating ambiguous business questions into robust analytical methods, validated outputs and clear insight narratives.

AI VALUE PROPOSITION

- Gen AI and AI agents focus: practical understanding of Gen AI concepts, AI architecture and design principles, with interest in applying AI agents to improve workflow automation, data validation and insight generation.
- Machine learning and statistical modelling: predictive modelling, classification, clustering, dimensionality reduction, time series forecasting, Bayesian inference, econometrics, multivariate analysis and model interpretation.
- Snowflake and analytics engineering: scalable, reproducible workflows in Snowflake, SQL, dbt, R and Python, with emphasis on data quality, validation logic, automation and documentation.
- Applied AI/NLP capability: Python ML libraries including PyTorch, scikit-learn, NumPy, pandas and Transformers, with experience in model training, evaluation and error analysis.
- Business impact: strong background converting ambiguous business questions into analytical methodology, tested logic, measurable outputs and stakeholder-ready insight narratives.

TECHNICAL SKILLS

Priority AI Keywords: Gen AI, AI Agents, Snowflake, Applied AI, Machine Learning, AI/NLP, Automation, Predictive Modelling.

Programming: Python, R, SQL, SAS Enterprise Miner, MATLAB, VBA.

Python Libraries: pandas, NumPy, scikit-learn, PyTorch, Transformers, Matplotlib.

Data Platforms: Snowflake, dbt Studio, Qlik, SharePoint, Jupyter Notebook, VS Code.

Analytics Methods: Predictive modelling, machine learning, time series, econometrics, Bayesian inference, multivariate analysis, data mining, anomaly/exception testing, financial modelling and quantitative risk analysis.

EDUCATION

Postgraduate Artificial Intelligence Coursework — University of Auckland: Completed COMPSCI 713 AI Fundamentals and COMPSCI 714 AI Architecture and Design, achieving grades of A- and B+ respectively. Coursework covered core AI concepts, architecture, design principles and applied AI foundations, 2026.

PhD, Applied Statistics — Auckland University of Technology, New Zealand: Full Scholarship, 2023. Research areas included market risk, asymmetric GARCH, R-copula, Bayesian MCMC, variational Bayes and ML optimisation.

MSc, Financial Mathematics — University of Exeter, United Kingdom: Distinction, 2014.

Master of Economics — Kasetsart University, Thailand: GPA 3.71/4.00, 2011.

BSc, Applied Mathematics — King Mongkut's University of Technology Thonburi, Thailand: 80th percentile, 2007.

Snowflake Hands-on Essentials — Verified Credential.

dbt Fundamentals — Verified Credential.

PROFESSIONAL EXPERIENCE

Data Specialist | 09/2025 – Present

Data Analyst | 09/2023 – 09/2025

Health New Zealand — Te Whatu Ora | Audit & Assurance Group, Computer Assisted Audit Tools and Techniques Team | Auckland, New Zealand

- Lead complex, high-volume analytics projects, ensuring robust methodology, quality assurance and defensible outcomes.

- Build scalable analytics workflows in R and Snowflake to process, validate and analyse large-scale operational and invoice datasets for audit and assurance functions.
- Develop automated testing programs and data pipelines to identify governance and conflict-of-interest risk indicators by matching declarations against supplier records using UiPath, Qlik and analytical rules.
- Translate business requirements into analytical logic, validation rules, exception tests and interactive dashboards in collaboration with auditors, business analysts and senior stakeholders.
- Present analytics functions, methodologies and insights to internal and external stakeholders, including senior leadership, Public Sector Assurance Forum and Audit New Zealand.
- Delivered evidence-based analysis using R, Snowflake, ACL and advanced Excel to support audit assurance, risk mitigation and governance reporting.
- Designed and implemented exception-testing frameworks for Holiday Act Remediation payments, improving discrepancy detection, payment accuracy assurance and transparency of reporting outputs.
- Improved repeatability and efficiency of analytics delivery through automated testing logic, standardised outputs, data quality checks and clear technical documentation.

Unit Pricing Analyst / 12/2022 – 09/2023

APEX Group / MMC — Fund Administration | Auckland, New Zealand

- Prepared and evaluated daily unit prices for client investment funds, applying reconciliation, analytical checks and issue investigation to support accurate financial reporting.
- Reviewed daily reconciliations, investigated exceptions and cleared differences using advanced Excel and structured analytical workflows.
- Identified and documented process improvements to reduce errors and cycle time using Excel macros and automation techniques.

Adjunct Lecturer — Data Analytics / 02/2022 – 12/2022

Massey University | New Zealand

- Delivered postgraduate Data Mining and Multivariate Analysis courses using SAS Enterprise Miner and R, including assessment design, grading and project supervision.
- Taught applied machine learning and statistical methods including logistic regression, decision trees, ensemble methods, Naïve Bayes, neural networks, SVMs, PCA, LDA, clustering and association analysis.
- Mentored students on analytics projects, model interpretation, research design and technical communication.

Tutorial Lecturer — Statistics / 09/2018 – 02/2022

Auckland University of Technology | Auckland, New Zealand

- Delivered applied statistics and mathematics tutorials to engineering and computing students, using R to support practical data analysis and modelling applications.
- Marked coursework and examinations and provided constructive feedback to enhance student performance and analytical capability.

Policy Researcher / 02/2015 – 09/2019

Fiscal Policy Research Institute Foundation, Ministry of Finance, Thailand; Thailand Development Research Institute | Thailand

- Conducted quantitative and qualitative policy research, integrating statistical analysis with market, industry and stakeholder insights.
- Developed technical reports outlining research scope, methodology and analytical framework, and presented interim and final findings to clients and expert stakeholders.
- Led fieldwork, stakeholder interviews, expert hearings and cross-functional coordination to support evidence-based recommendations.

Bond Pricing Analyst / 11/2014 – 01/2015; 10/2015 – 05/2017

Thai Bond Market Association | Thailand

- Provided daily Thai bond mark-to-market prices and developed methodologies for pricing new bonds and structured products using VBA Excel and MATLAB.
- Conducted quantitative research on financial markets, bond pricing conventions, money market development and data services.
- Delivered bond pricing analysis workshops for market participants, translating technical pricing methodologies into practical guidance.

Software Developer / 12/2008 – 11/2012

DST Worldwide Services | Kasikorn Bank | AIA Thailand | Thailand

- Designed, developed and maintained enterprise applications using COBOL, VSAM and SQL in banking and financial systems environments.
- Produced technical specifications, wrote maintainable code and completed unit testing to support stable QA handover and system performance.

CERTIFICATIONS, SPEAKING ENGAGEMENTS & PUBLICATIONS

- Invited Keynote Speaker, IIA NZ Conference 2024, Museum of New Zealand Te Papa, Wellington — Harnessing Data Analytics: Building Resilience for Internal Auditors in Turbulent Times.
- Invited Speaker, Institute of Internal Auditors Education Day 2024, KPMG Auckland — Data Analytics and AI.

- Publication: Bayesian Estimation of R-vine Copula with Gaussian-Mixture GARCH Margins: An MCMC and Machine Learning Comparison, Mathematics.
- Publication: Modelling and Forecasting COVID-19 Stock Returns Using Asymmetric GARCH-ICAPM with Mixture and Heavy-Tailed Distributions, Applied Economics.
- Publication: Analysis of Nonlinear Comovement of Benchmark Thai Government Bond Yields, PIER Discussion Papers.

ADDITIONAL INFORMATION

Languages: English and Thai.

Work authorisation: New Zealand Permanent Resident.